

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: from google.colab import files
uploaded = files.upload()
```

Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.

Saving telco_churn.csv to telco_churn.csv

```
In [3]: df = pd.read_csv("telco_churn.csv")
df
```

```
Out[3]:
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	Mu
0	7590-VHVEG	Female	0	Yes	No	1	No	
1	5575-GNVDE	Male	0	No	No	34	Yes	
2	3668-QPYBK	Male	0	No	No	2	Yes	
3	7795-CFOCW	Male	0	No	No	45	No	
4	9237-HQITU	Female	0	No	No	2	Yes	
...	...	...	...	...	...	...	...	...
7038	6840-RESVB	Male	0	Yes	Yes	24	Yes	
7039	2234-XADUH	Female	0	Yes	Yes	72	Yes	
7040	4801-JJAZL	Female	0	Yes	Yes	11	No	
7041	8361-LTMKD	Male	1	Yes	No	4	Yes	
7042	3186-AJIEK	Male	0	No	No	66	Yes	

7043 rows × 21 columns

```
In [4]: df.shape
```

```
Out[4]: (7043, 21)
```

```
In [5]: df.columns
```

```
Out[5]: Index(['customerID', 'gender', 'SeniorCitizen', 'Partner', 'Dependents',
              'tenure', 'PhoneService', 'MultipleLines', 'InternetService',
              'OnlineSecurity', 'OnlineBackup', 'DeviceProtection', 'TechSupport',
              'StreamingTV', 'StreamingMovies', 'Contract', 'PaperlessBilling',
              'PaymentMethod', 'MonthlyCharges', 'TotalCharges', 'Churn'],
              dtype='object')
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
#   Column                Non-Null Count  Dtype
---  -
0   customerID            7043 non-null   object
1   gender                 7043 non-null   object
2   SeniorCitizen          7043 non-null   int64
3   Partner                7043 non-null   object
4   Dependents             7043 non-null   object
5   tenure                 7043 non-null   int64
6   PhoneService           7043 non-null   object
7   MultipleLines          7043 non-null   object
8   InternetService        7043 non-null   object
9   OnlineSecurity         7043 non-null   object
10  OnlineBackup           7043 non-null   object
11  DeviceProtection       7043 non-null   object
12  TechSupport            7043 non-null   object
13  StreamingTV            7043 non-null   object
14  StreamingMovies        7043 non-null   object
15  Contract               7043 non-null   object
16  PaperlessBilling       7043 non-null   object
17  PaymentMethod          7043 non-null   object
18  MonthlyCharges         7043 non-null   float64
19  TotalCharges           7043 non-null   object
20  Churn                  7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
```

```
In [8]: df.isnull().sum()
```

Out[8]: 0

customerID	0
gender	0
SeniorCitizen	0
Partner	0
Dependents	0
tenure	0
PhoneService	0
MultipleLines	0
InternetService	0
OnlineSecurity	0
OnlineBackup	0
DeviceProtection	0
TechSupport	0
StreamingTV	0
StreamingMovies	0
Contract	0
PaperlessBilling	0
PaymentMethod	0
MonthlyCharges	0
TotalCharges	0
Churn	0

dtype: int64

```
In [9]: # DATA CLEANING
# CONVERTING TOTAL CHARGES INTO NUMERIC COLUMN
df['TotalCharges'] = pd.to_numeric(df['TotalCharges'], errors='coerce')
```

```
In [10]: # HANDLE MISSING VALUES
df.isnull().sum()
df.dropna(inplace = True)
```

```
In [11]: # TARGET COLUMN
df['Churn'].value_counts()
```

Out[11]:

count	
Churn	
No	5163
Yes	1869

dtype: int64

```
In [12]: # CONVERT TARGET COLUMN YES NO
df['Churn'] = df['Churn'].map({'Yes': 1, 'No': 0})
```

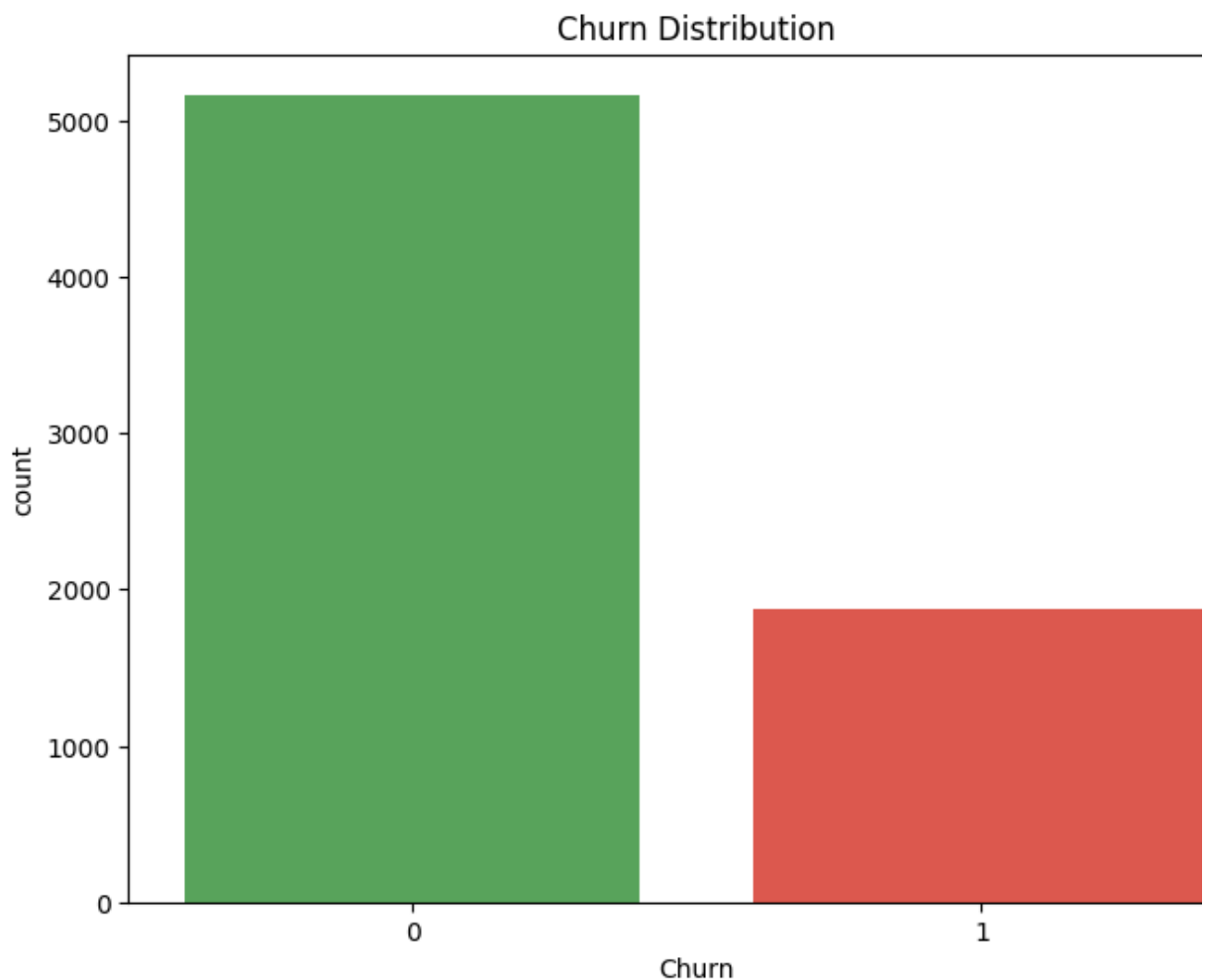
EXPLORATORY DATA ANALYSIS

```
In [15]: # 1 CHURN DISTRIBUTION
plt.figure(figsize=(8,6))
sns.countplot(x='Churn', data=df, palette=['#4CAF50', '#F44336'])
plt.title('Churn Distribution')
plt.show()
```

/tmp/ipykernel_11187/486553737.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effect.

```
sns.countplot(x='Churn', data=df, palette=['#4CAF50', '#F44336'])
```



Short Insights

~73% customers retained, ~27% churned

Around 1 in 4 customers is leaving

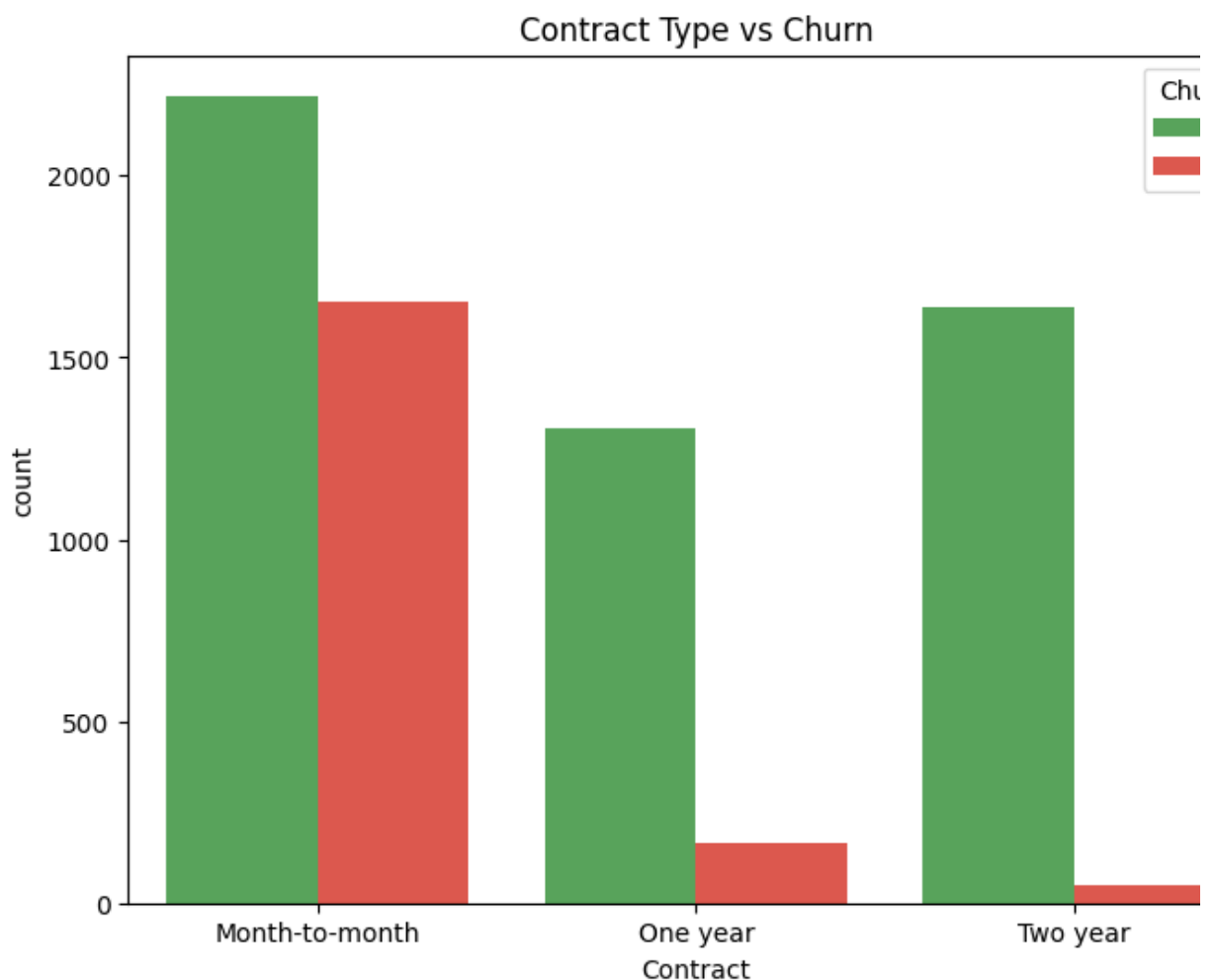
Dataset is imbalanced (more non-churn than churn)

Churn rate is significant → needs attention

Big opportunity: reducing churn can boost revenue

□ Focus next on: contract type, tenure, and charges to understand why customers churn.

```
In [16]: # 2 □ Contract Type vs Churn (MOST IMPORTANT)
plt.figure(figsize=(8, 6))
sns.countplot(x='Contract', hue='Churn', data=df, palette=['#4CAF50', '#F44336'])
plt.title('Contract Type vs Churn')
plt.show()
```



□ Short Detailed Insights

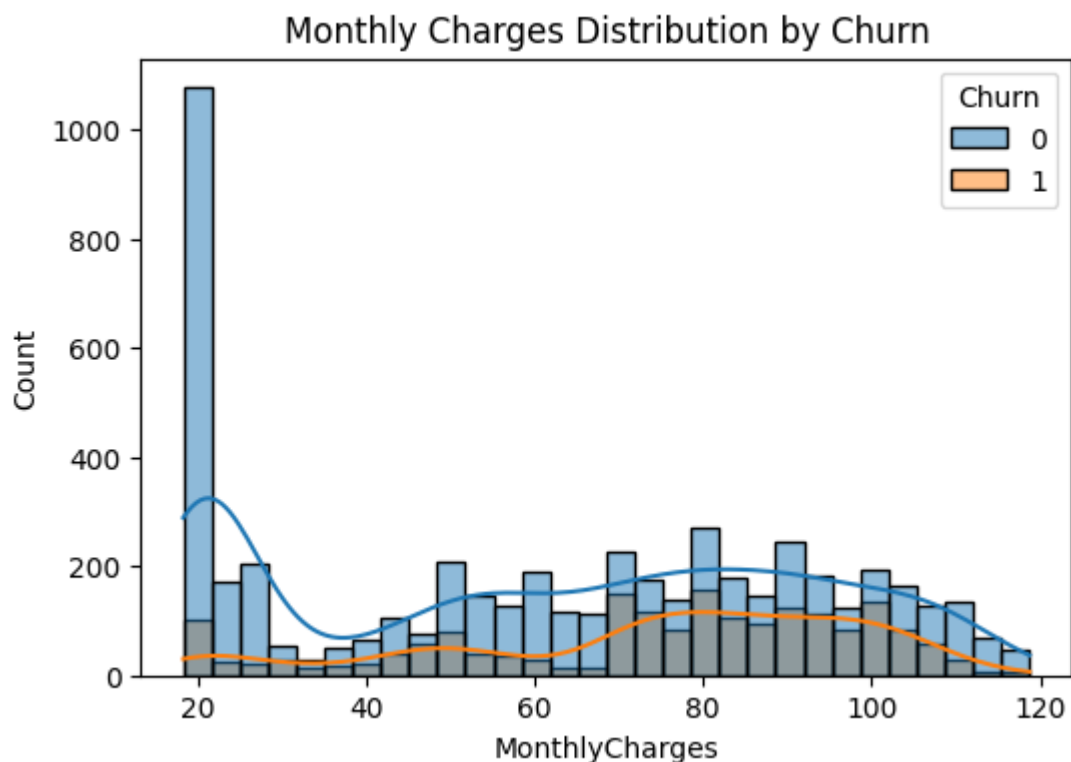
- **Month-to-month contracts have the highest churn** → Churn is almost as high as retent
These customers are **high-risk**

- **One-year contracts show much lower churn** → Majority customers are retained □Comm reduces churn
- **Two-year contracts have the lowest churn** → Almost **no customers leave** □Long-term c = **strong loyalty**
- **Clear pattern: Longer contract → Lower churn** □Contract duration is a **key driver of ch**

□ Business Takeaway

Encourage users to move from **monthly** → **yearly plans** to significantly reduce churn.

```
In [20]: # 3 monthly charges vs churn
plt.figure(figsize=(6,4))
sns.histplot(data=df, x='MonthlyCharges', hue='Churn', bins=30, kde=True)
plt.title("Monthly Charges Distribution by Churn")
plt.show()
```



□ Insights: Monthly Charges vs Churn

- **Higher charges → higher churn** □Churned customers are concentrated in the **₹70–₹110**
- **Low charges customers rarely churn** □Strong retention in **₹20–₹40 range**
- **Churn distribution is shifted right** □Customers who leave tend to pay **more than average**
- **Retention spread is wider** □Non-churn customers exist across all price ranges, especially ones
- **Price sensitivity is a key factor** □Expensive plans increase churn risk

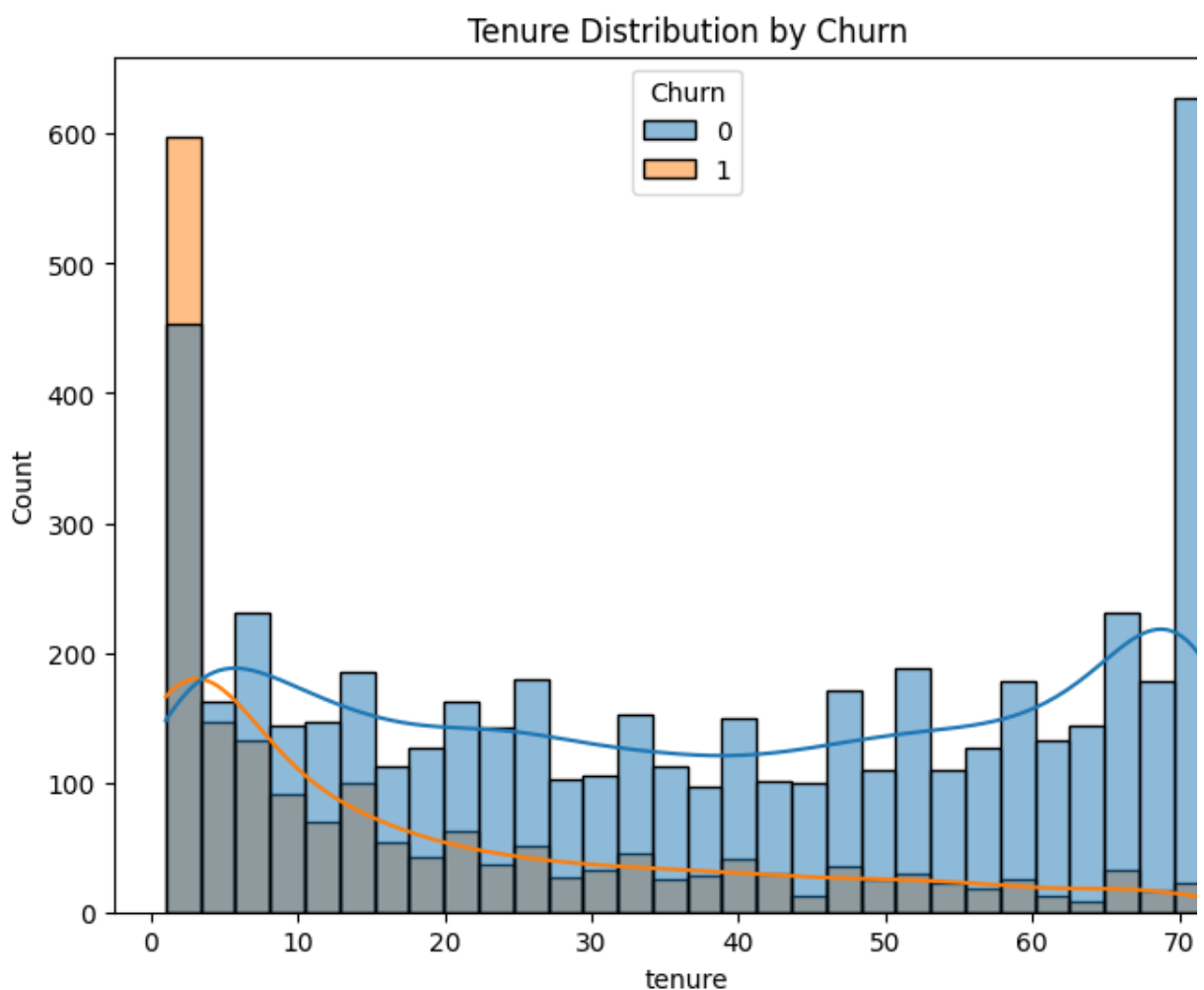
□ Business Insight

Customers paying higher monthly charges are more likely to leave → focus on:

- Discounts / offers
- Better value perception
- Premium customer experience

□ High-paying customers = **high churn risk + high revenue impact**

```
In [25]: # 4 □ Tenure vs Churn
plt.figure(figsize=(8,6))
sns.histplot(data=df, x='tenure', hue='Churn', bins=30, kde=True)
plt.title('Tenure Distribution by Churn')
plt.show()
```



□ Insights: Tenure vs Churn

- **New customers churn the most** □ High churn in **0–10 months tenure**
- **Churn decreases as tenure increases** □ Customers staying longer are **less likely to leave**
- **Long-tenure customers are highly loyal** □ Very low churn after **50+ months**
- **Retention peaks at high tenure** □ Majority of non-churn customers are in **60–72 months**

- Critical risk period = early stage □ First few months are **make-or-break**

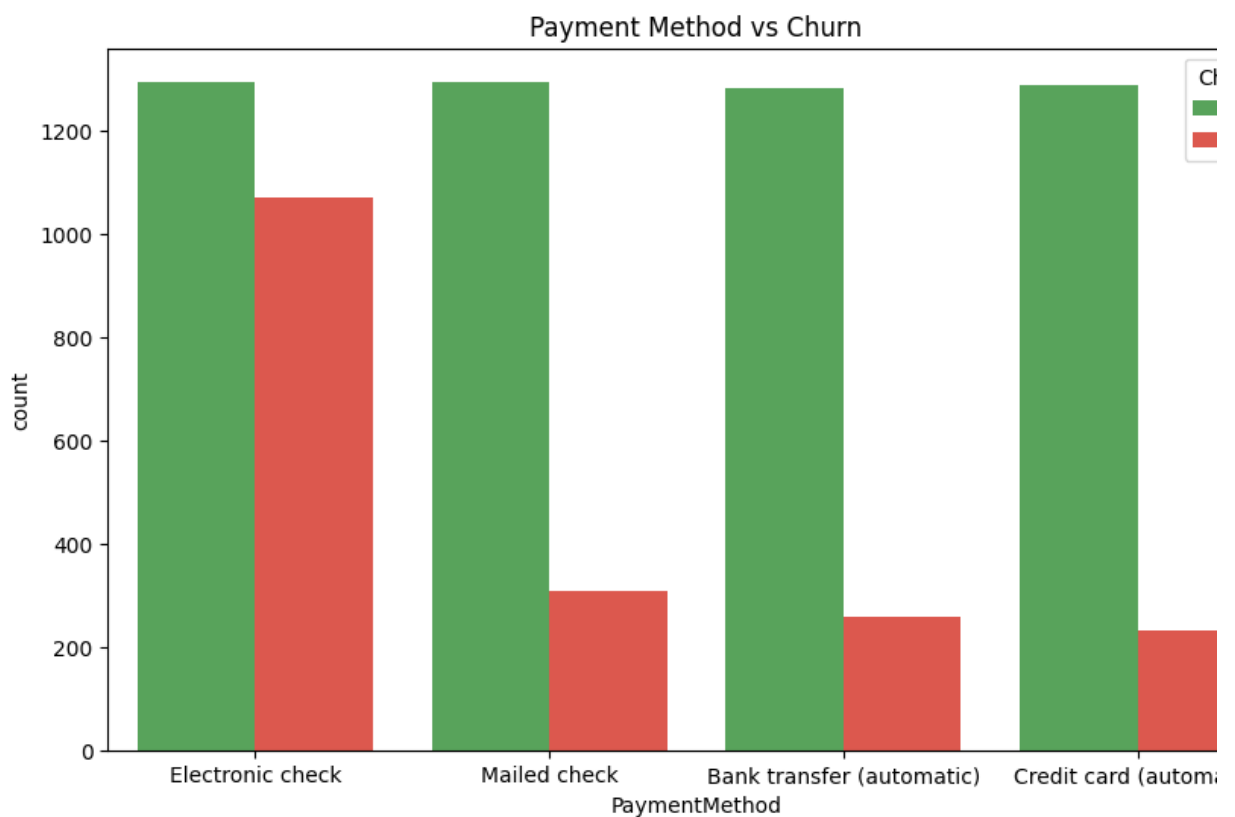
□ Business Insight

Focus on **early customer experience (first 3–6 months)**:

- Onboarding improvements
- Customer support
- Engagement strategies

□ Retain early → long-term loyal customers □

```
In [27]: # 5 payment method vs churn
plt.figure(figsize=(10,6))
sns.countplot(x='PaymentMethod', hue='Churn', data=df, palette=['#4CAF50', '#F44336'])
plt.title('Payment Method vs Churn')
plt.show()
```



□ Insights: Payment Method vs Churn

- **Electronic check has the highest churn** □ Churn is **very high compared to other methods**
Major risk segment
 - **Automatic payments have lowest churn** □ *Bank transfer & credit card (auto)* show **strong retention** □ Auto-pay users are more stable
 - **Mailed check has moderate churn** □ Better than electronic check but worse than auto-pay
 - **Clear pattern: Manual payment → Higher churn** □ Automatic payment → **lower churn**
-

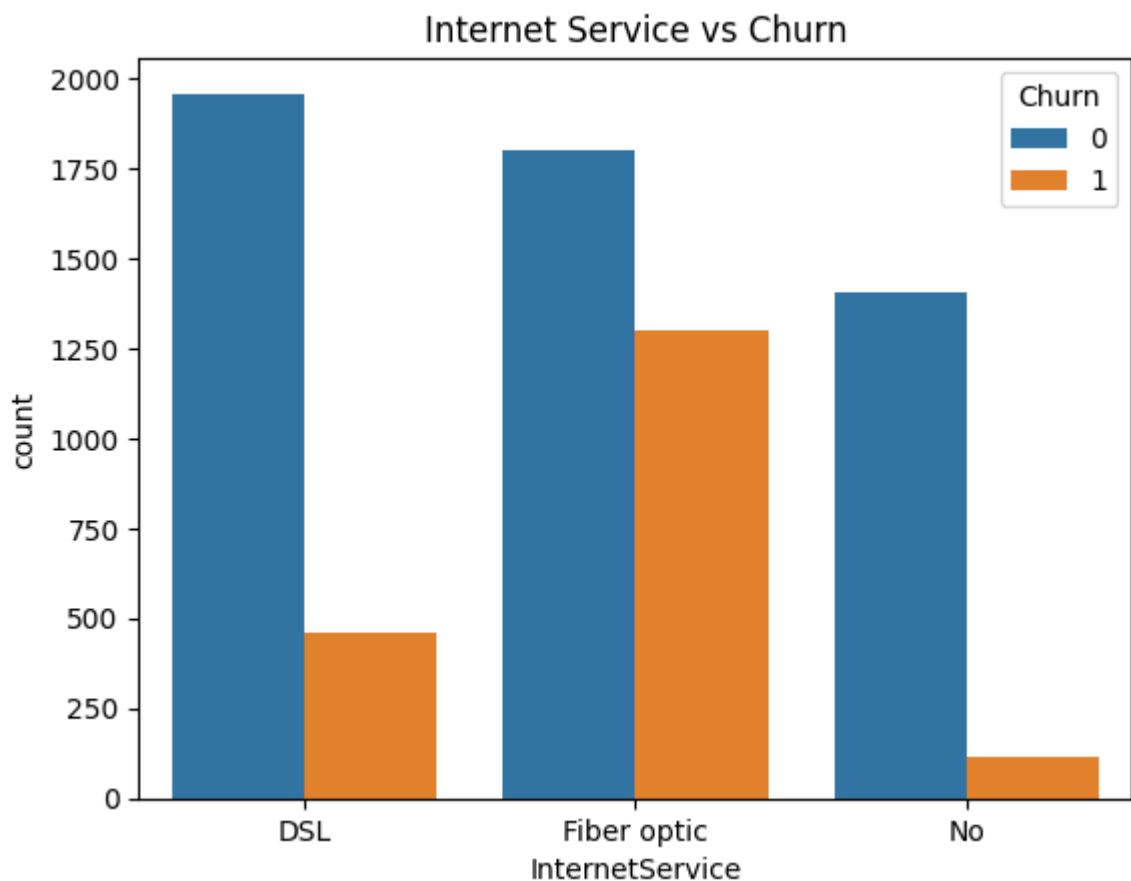
□ Business Insight

Encourage customers to switch to **auto-payment methods**:

- Offer discounts for auto-pay
- Simplify payment setup

□ Payment behavior is a **strong churn indicator** □

```
In [29]: # 6 INTERNET SERVICE VS CHURN
sns.countplot(x='InternetService', hue='Churn', data=df)
plt.title("Internet Service vs Churn")
plt.show()
```



□ Insights: Internet Service vs Churn

- **Fiber optic users have the highest churn** □ Churn is **very high compared to DSL and no internet service** □ High-value but **high-risk segment**
 - **DSL users show moderate churn** □ Better retention than fiber users □ More stable customer base
 - **Customers with no internet have lowest churn** □ Very few customers leave □ Possibly low expectations or basic usage
 - **Clear pattern: Better service ≠ lower churn** □ Fiber users may face **high cost or service quality issues**
-

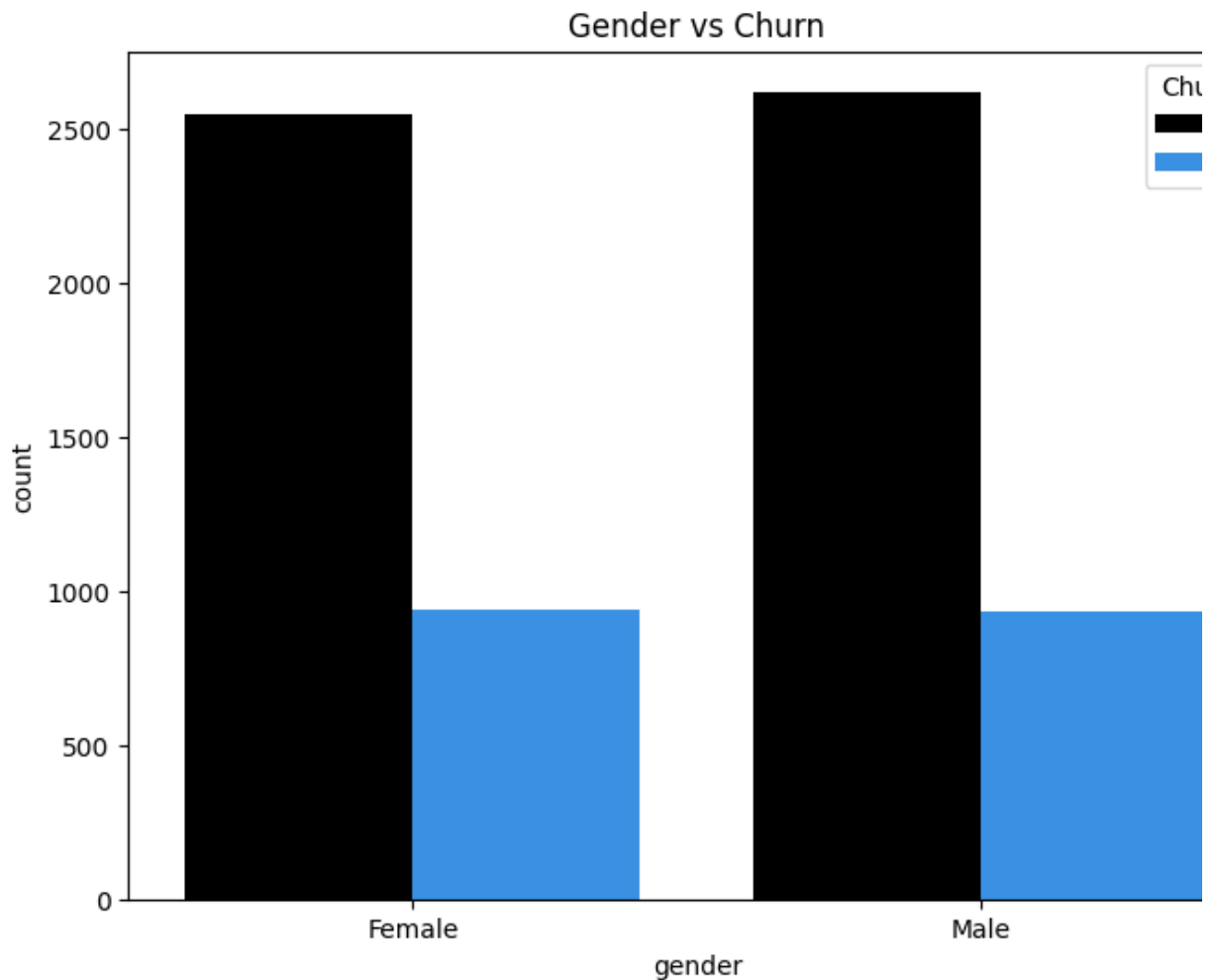
□ Business Insight

Focus on **fiber optic customers**:

- Improve service quality
- Address pricing concerns
- Provide better support

□ Retaining fiber users is critical since they are **high revenue but high churn** □

```
In [33]: # 7 GENDER VS CHURN
plt.figure(figsize=(8,6))
sns.countplot(x='gender', hue='Churn', data=df,
              palette=['#000000', '#1E90FF'])
plt.title('Gender vs Churn')
plt.show()
```



□ Insights: Gender vs Churn

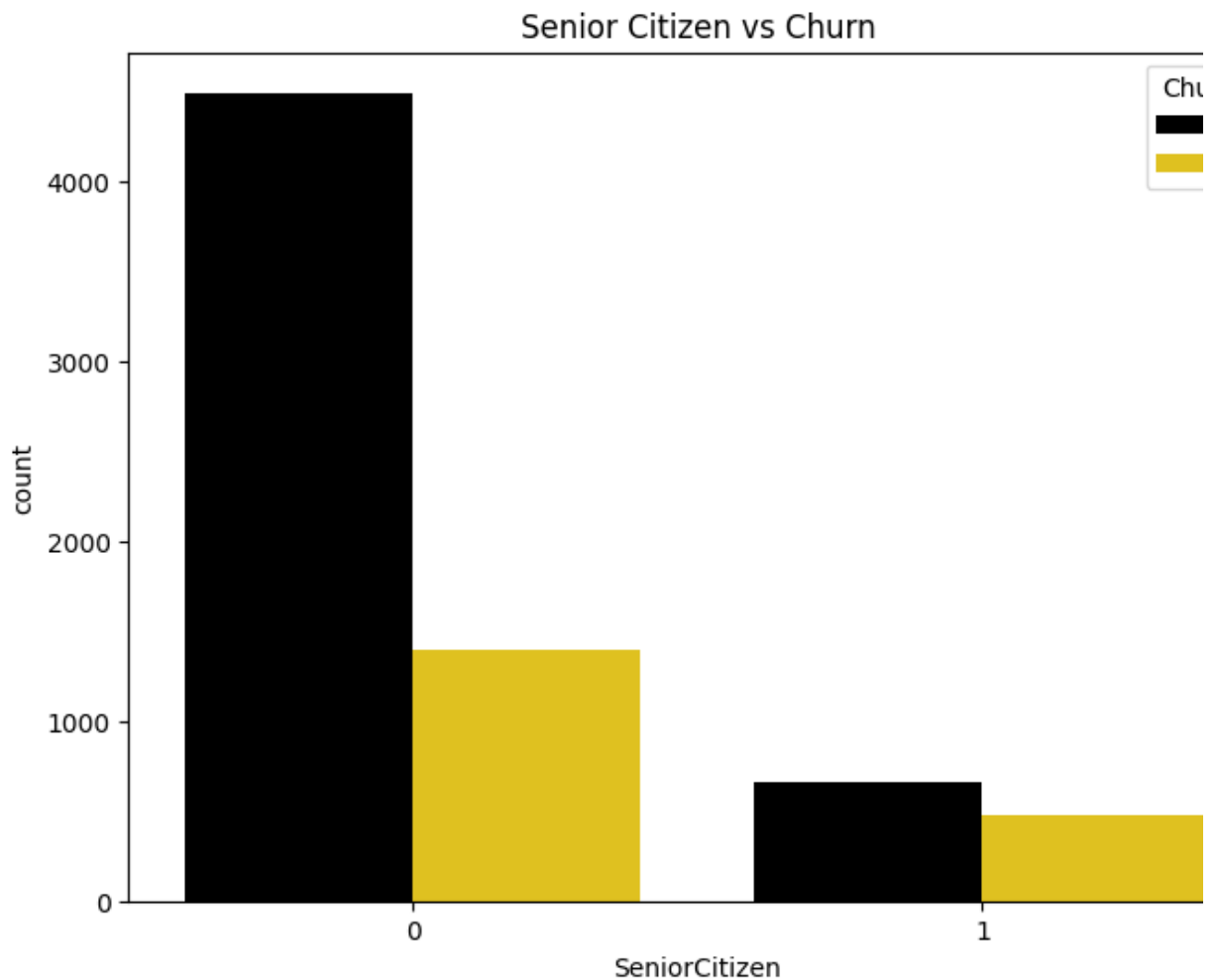
- **Churn is almost equal for both genders** □ Male and Female customers show **very similar** patterns
- **No strong gender impact** □ Gender is **not a key driver of churn**
- **Slightly higher male count** □ More male customers overall, but churn % remains similar
- **Retention pattern is consistent** □ Both genders have **similar retention behavior**

□ Business Insight

- Gender-based targeting is **not effective**
- Focus more on:
 - Contract type
 - Tenure
 - Monthly charges

□ These factors influence churn much more □

```
# 8 Senior Citizen vs Churn
plt.figure(figsize=(8,6))
sns.countplot(x='SeniorCitizen', hue='Churn', data=df,
              palette=['#000000', '#FFD700'])
plt.title('Senior Citizen vs Churn')
plt.show()
```



□ Insights: Senior Citizen vs Churn

- **Senior citizens have higher churn rate** □ Proportion of churn is **much higher for senior** (1) compared to non-senior (0)
- **Non-senior customers dominate overall** □ Majority customers are non-senior, but their churn is lower
- **Senior segment is high-risk** □ Even with fewer customers, churn is **relatively high**
- **Retention is stronger in non-senior group** □ Most non-senior customers stay longer

□ Business Insight

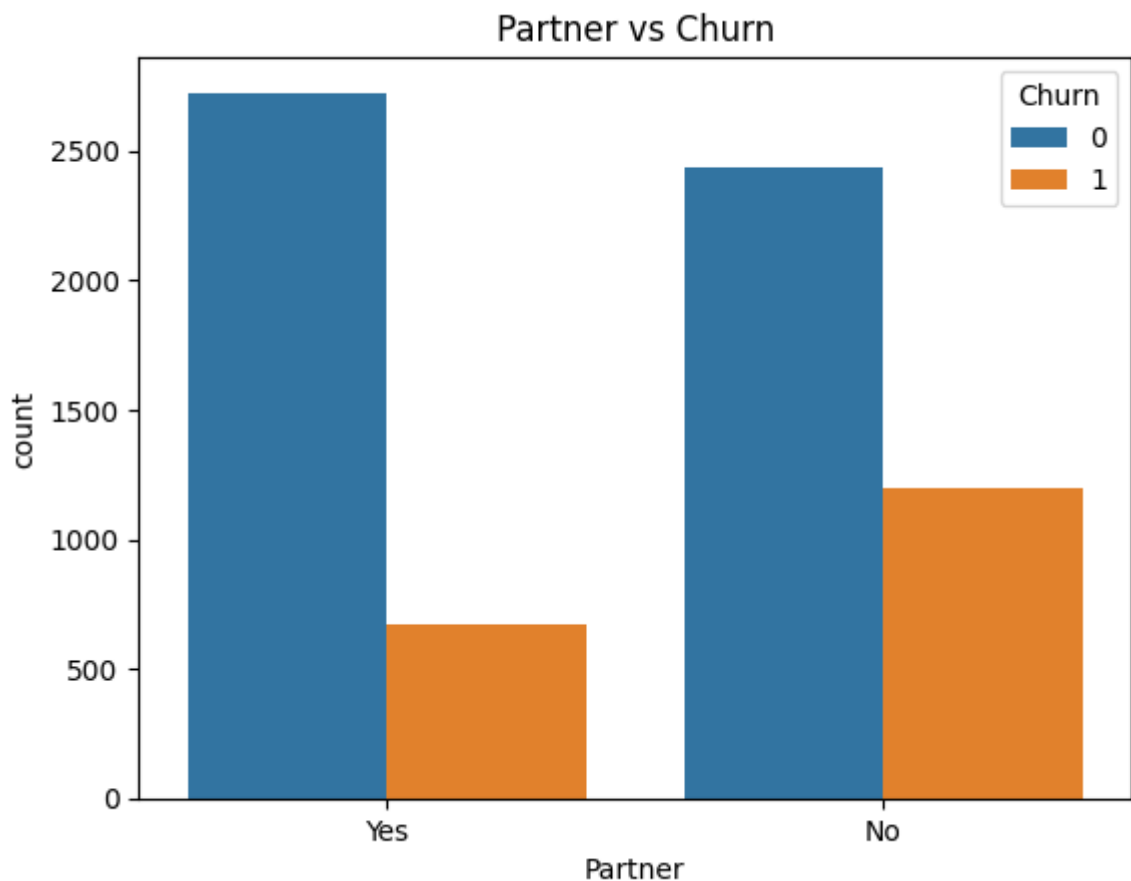
Focus on **senior citizens**:

- Better customer support
- Simpler plans/pricing

- Personalized assistance

□ Senior customers = **smaller group but higher churn risk** □

```
In [36]: # 9 PARTNER VS CHURN
sns.countplot(x='Partner', hue='Churn', data=df)
plt.title("Partner vs Churn")
plt.show()
```



□ Insights: Partner vs Churn

- **Customers without a partner churn more** □ “No partner” group shows **significantly higher** churn rate
- **Customers with a partner are more loyal** □ Higher retention in **partner = Yes**
- **Partner status impacts stability** □ Having a partner likely indicates **more stable usage/loyalty** and **commitment**
- **Clear behavioral pattern** □ Single customers = **higher churn risk**

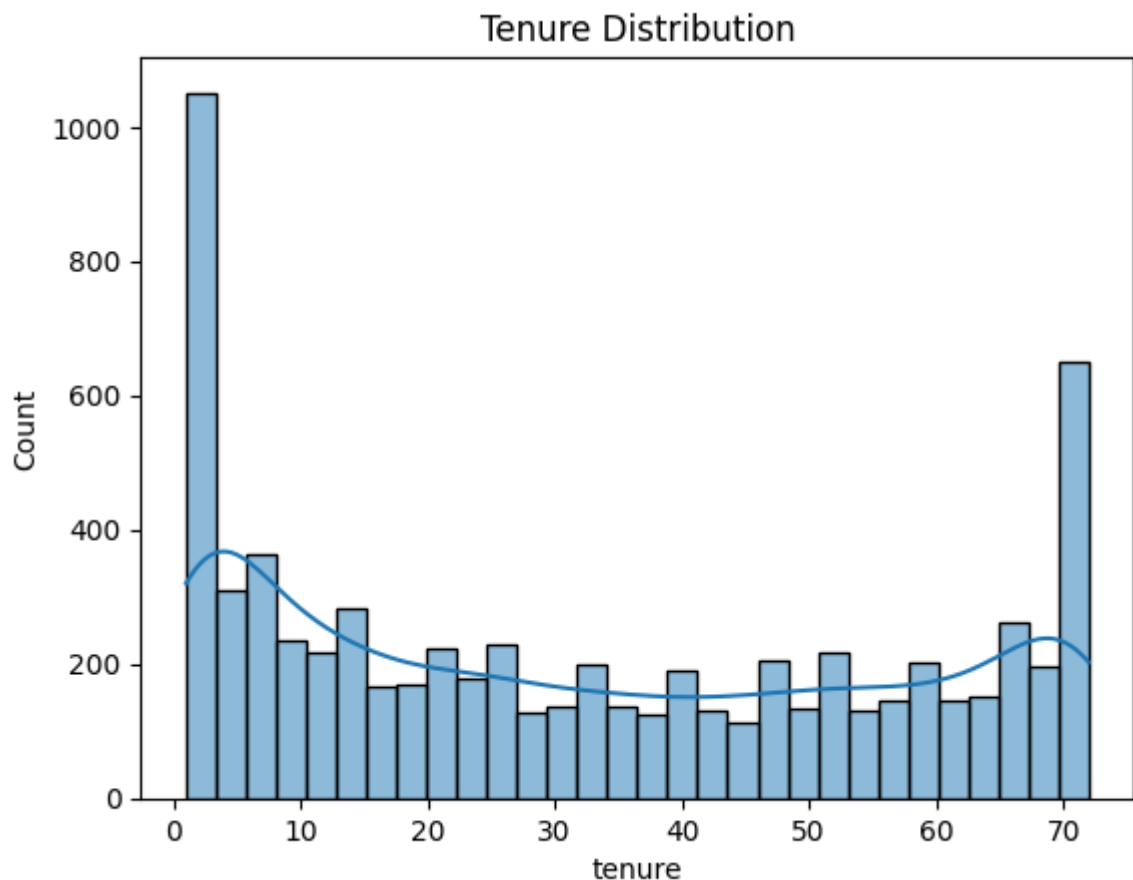
□ Business Insight

Target customers without partners:

- Personalized offers
- Engagement campaigns
- Flexible plans

□ “No partner” segment = **key churn risk group** □

```
In [42]: # 10 TENURE DISTRIBUTION
sns.histplot(df['tenure'], bins=30, kde=True)
plt.title("Tenure Distribution")
plt.show()
```



□ Insights: Tenure Distribution

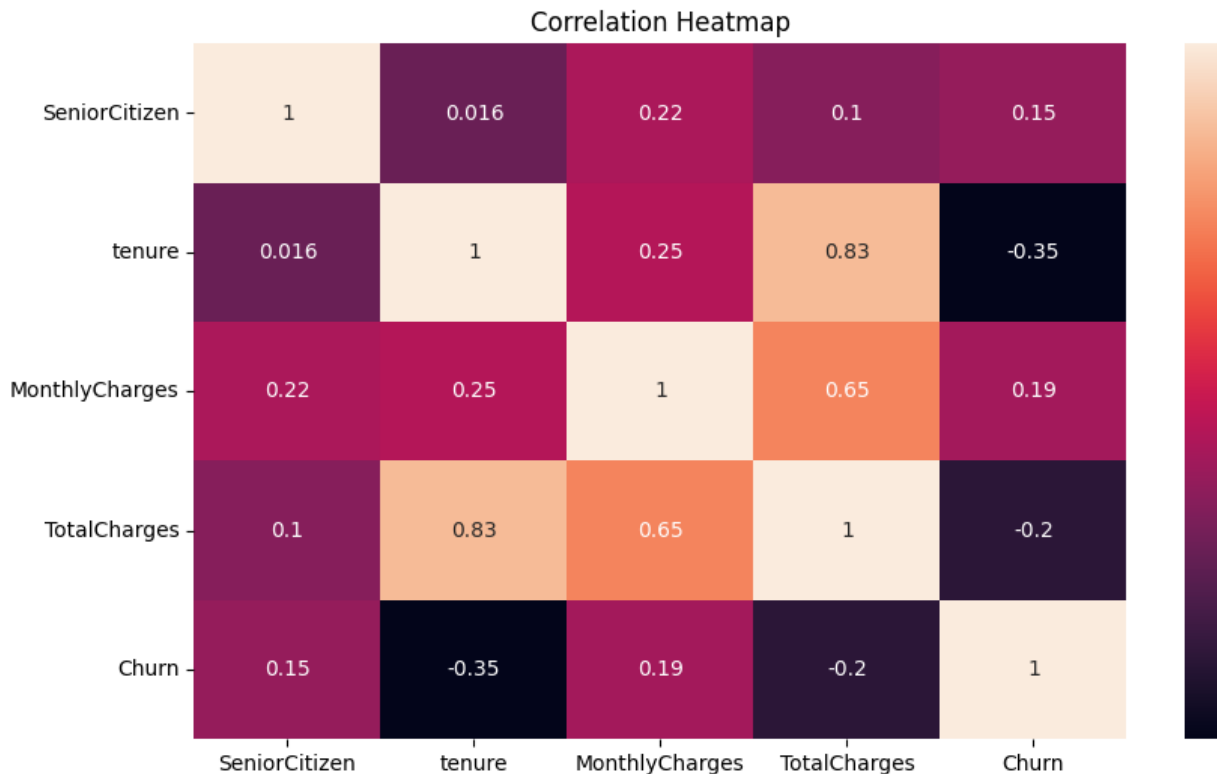
- **High number of new customers (0–5 months)** □ Indicates strong acquisition but also **early risk**
- **Spike at very high tenure (65–72 months)** □ Many **long-term loyal customers**
- **Middle tenure (10–50 months) is relatively stable** □ No extreme peaks → steady customer base
- **U-shaped pattern** □ More customers at **start and end**, fewer in the middle
- **Customer lifecycle is clear** □ Early stage (risky) → mid stage (stable) → long-term (loyal)

□ Business Insight

- Focus on **early retention (first few months)**
- Maintain engagement to move users into **long-term loyal segment**

□ Goal: Convert new users → long-term customers □

```
In [40]: # 11 CORRELATION HEATMAP
plt.figure(figsize=(10,6))
sns.heatmap(df.corr(numeric_only=True), annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



□ Insights: Correlation Heatmap

- **Tenure has strongest negative correlation with churn (-0.35)** □ Longer stay → **less likely to churn** (very important feature)
- **Monthly Charges positively correlated with churn (0.19)** □ Higher charges → **higher churn**
- **Total Charges negatively correlated with churn (-0.20)** □ High total spenders are **more likely to stay**
- **Strong correlation: Tenure vs Total Charges (0.83)** □ Logical: longer customers → **more likely to spend**
- **Monthly vs Total Charges (0.65)** □ Higher monthly plans → higher total revenue
- **Senior Citizen has weak correlation (0.15)** □ Slight impact but **not a strong driver**

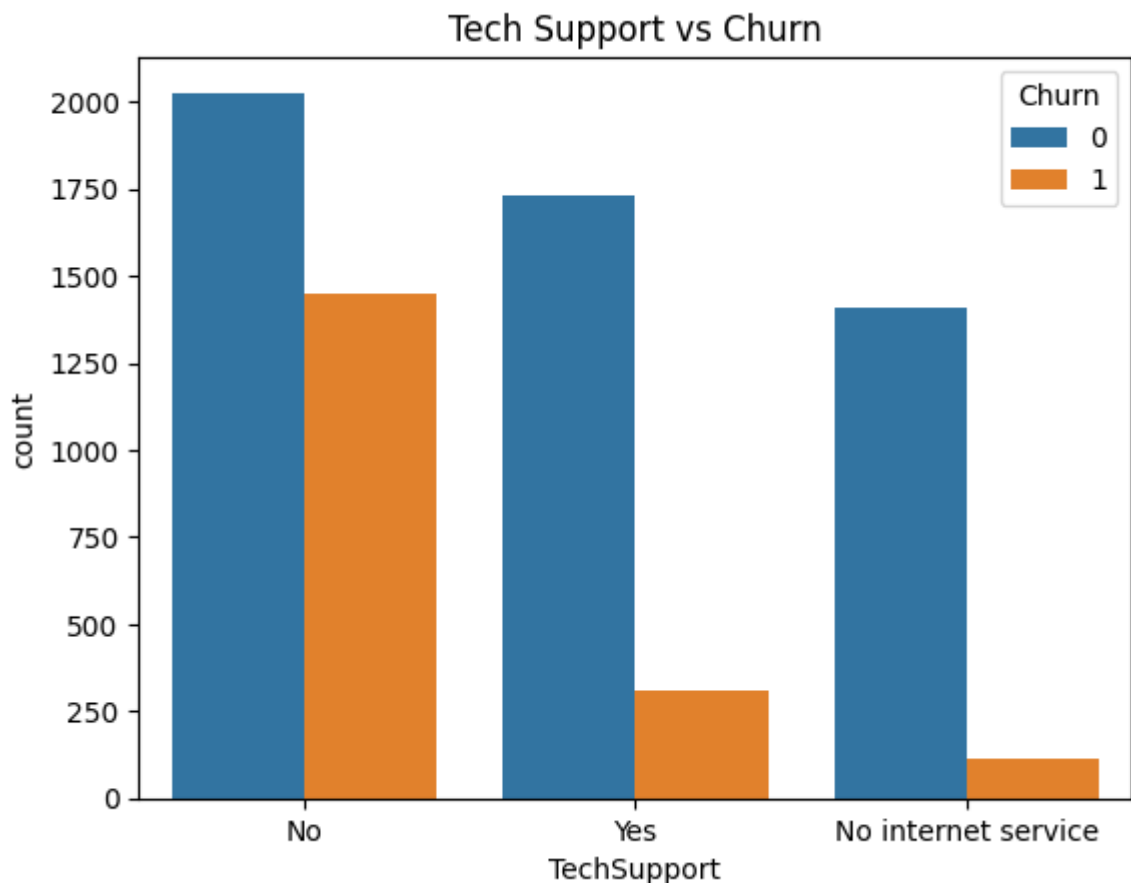
□ Business Insight

- **Top churn drivers:** Tenure (↓), Monthly Charges (↑)
- Focus on:
 - Retaining **new customers (low tenure)**

- Managing **high monthly cost perception**

□ Tenure is the **most important feature for churn prediction** □

```
In [43]: # 12 TECH SUPPORT VS CHURN
sns.countplot(x='TechSupport', hue='Churn', data=df)
plt.title("Tech Support vs Churn")
plt.show()
```



□ Insights: Tech Support vs Churn

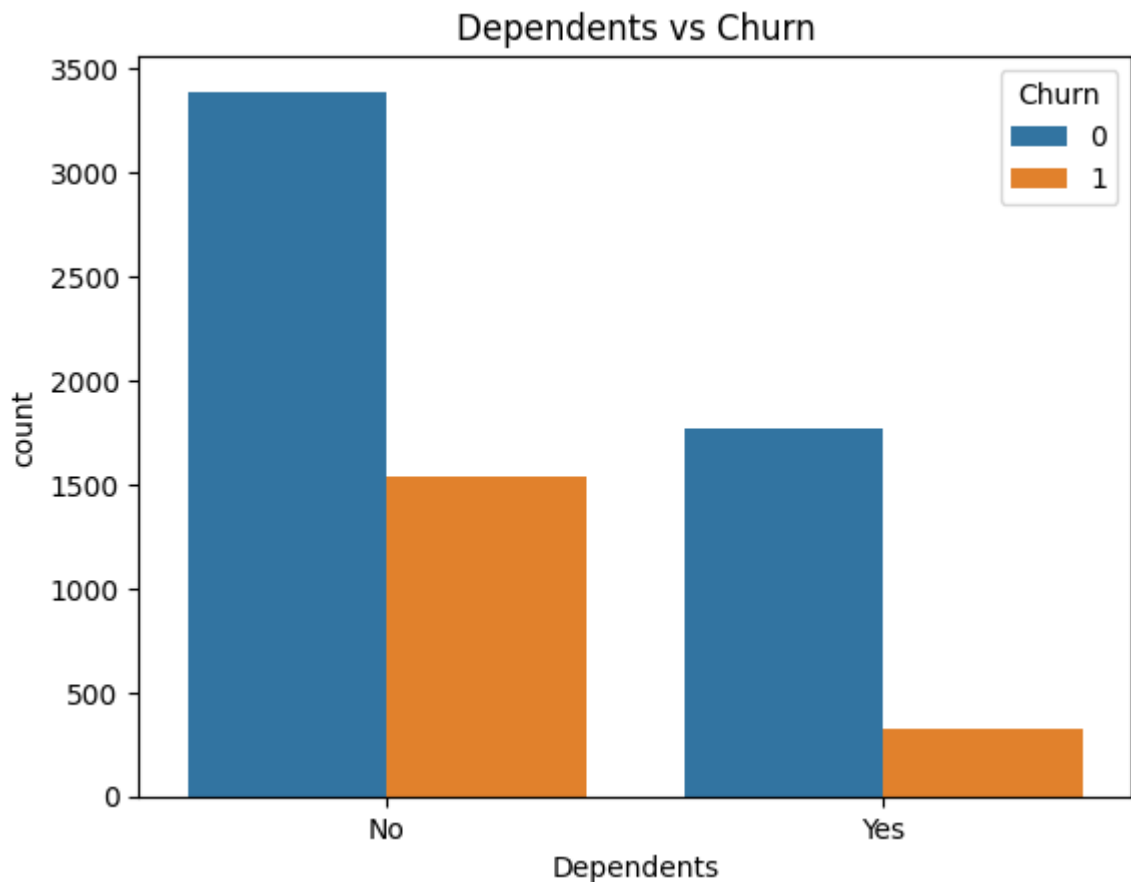
- **No tech support** → **highest churn** □ Customers without support show **very high churn**
- **Having tech support reduces churn significantly** □ “Yes” group has **much better retention**
- **No internet service** → **lowest churn** □ Minimal churn, likely low dependency users
- **Support availability impacts loyalty** □ Customers with help available are **more satisfied & loyal**

□ Business Insight

- Promote **tech support services**
- Offer **free trials / bundled support plans**
- Improve **customer issue resolution**

□ Tech support = **strong churn reduction factor** □

```
In [44]: # 13 DEPENDENTS VS CHURN
sns.countplot(x='Dependents', hue='Churn', data=df)
plt.title("Dependents vs Churn")
plt.show()
```



□ Insights: Dependents vs Churn

- **Customers without dependents churn more** □ “No dependents” group shows **higher churn**
- **Customers with dependents are more loyal** □ Lower churn in **dependents = Yes**
- **Family responsibility = stability** □ Customers with dependents likely prefer **long-term contracts**
- **Clear pattern** □ No dependents → **higher churn risk**

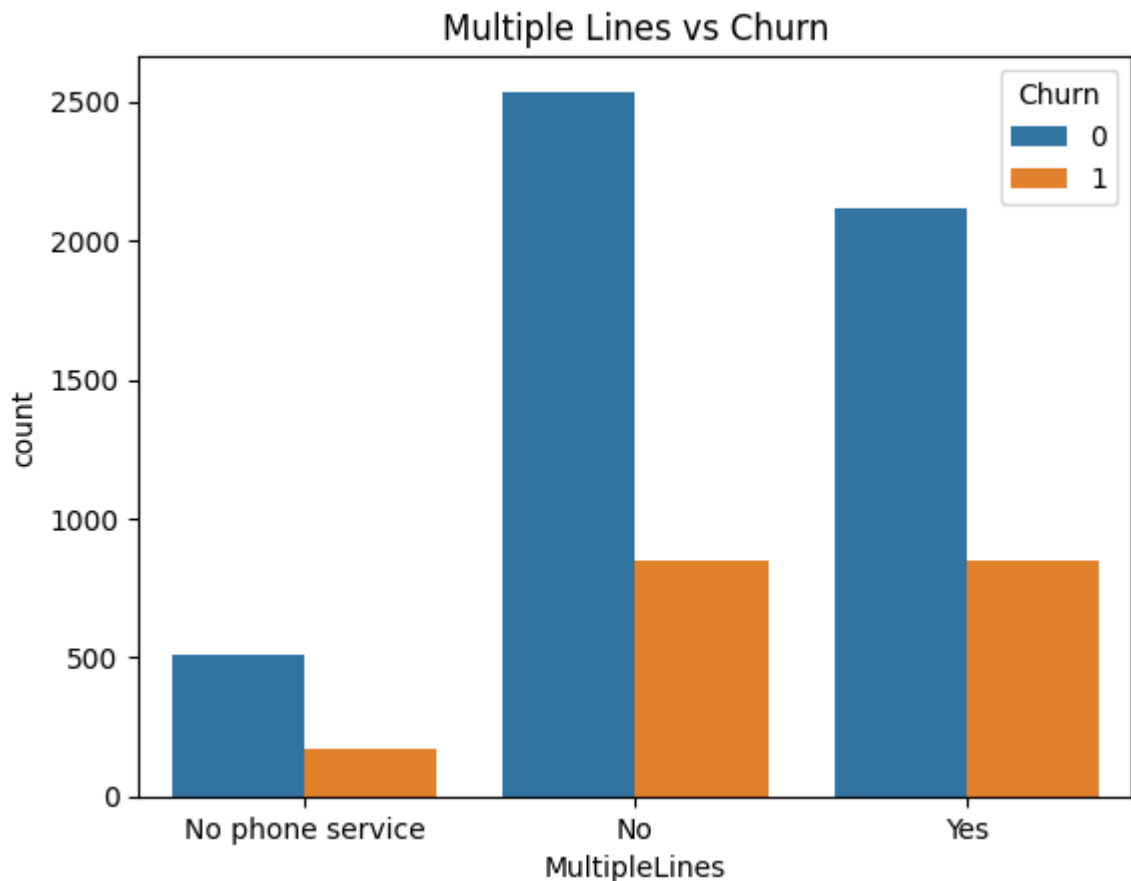
□ Business Insight

- Target customers without dependents:
 - Engagement offers
 - Flexible plans
 - Personalized deals

□ Customers with dependents = **more stable segment** □

```
In [45]: # 14 MULTIPLE LINES VS CHURN
sns.countplot(x='MultipleLines', hue='Churn', data=df)
```

```
plt.title("Multiple Lines vs Churn")
plt.show()
```



□ Insights: Multiple Lines vs Churn

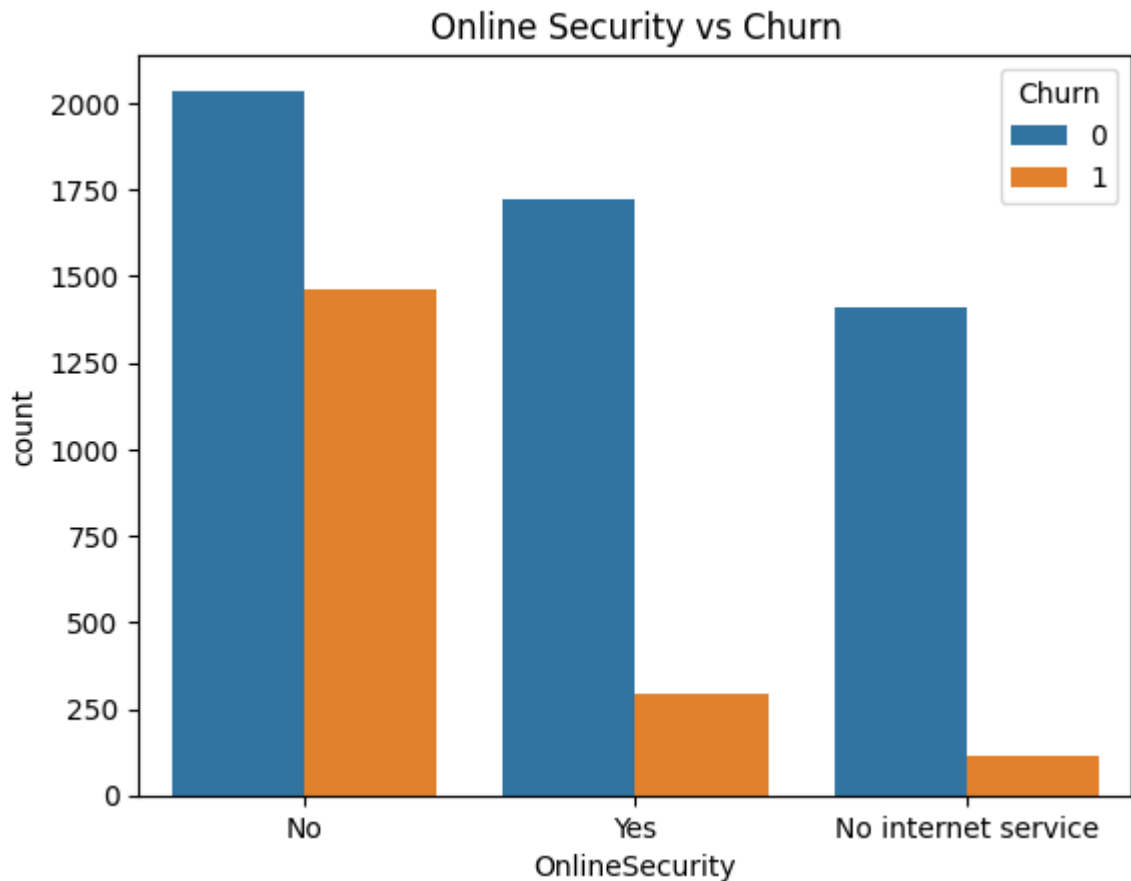
- **Customers without multiple lines churn slightly more** □ “No” group shows **higher churn** “Yes”
- **Multiple lines users are a bit more stable** □ Having multiple connections → **lower churn tendency**
- **No phone service** → **low churn** □ Small segment, less impact on overall churn
- **Weak to moderate impact feature** □ Difference exists but **not very strong**

□ Business Insight

- Upsell customers to **multiple lines / bundled plans**
- Create **family or combo offers**

□ Bundled services = **better retention & customer stickiness** □

```
In [46]: # 15 ONLINE SECURITY VS CHURN
sns.countplot(x='OnlineSecurity', hue='Churn', data=df)
plt.title("Online Security vs Churn")
plt.show()
```



□ Insights: Online Security vs Churn

- **No online security → highest churn** □ Customers without security show **very high churn**
 - **Having online security reduces churn sharply** □ “Yes” group has **strong retention**
 - **No internet service → very low churn** □ Small, stable segment
 - **Security adds customer confidence** □ Users with protection feel **safer and more satisfied**
 - **Strong churn driver** □ Online security is a **key factor influencing churn**
-

□ Business Insight

- Bundle **online security with plans**
- Offer **free trials / discounts**
- Educate users about benefits

□ Security services = **powerful churn reduction strategy** □